iLearn Practice Exam

11BR · Chapter / Topic 8

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Topic 8: Databases

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 73

Marks obtained: ..... / 73

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 73.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 5 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

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1 A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

CUSTOMER(CustomerID, CustomerName, Email)

ORDER(OrderID, CustomerID, Date, Collected)

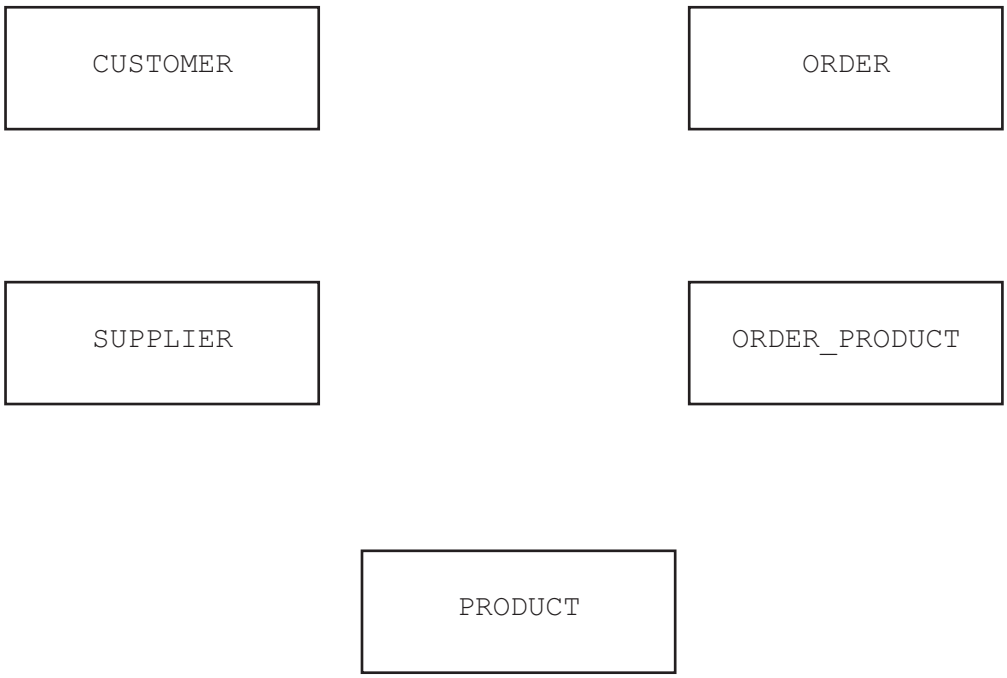
ORDER_PRODUCT(OrderID, ProductID, Quantity)

PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)

SUPPLIER(SupplierID, SupplierName, SupplierEmail)

The primary keys are underlined.

(a) Complete the entity-relationship (E-R) diagram for the database.



[4]

(b) A new product needs to be entered into the database. The product has the ID 002323, the product name 'Blue ball point 2mm', there are 50 in a box, the product costs $5.00 and the supplier has the ID SFX223.

Write a Structured Query Language (SQL) script to enter the new product into the table `PRODUCT`.

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(c) The attribute `Collected` in the table `ORDER` stores the Boolean value `TRUE` if the order has been collected and `FALSE` if the order has not been collected.

Write an SQL script to return the customer name for each customer that has orders they have not collected. Include the number of orders each customer has not collected with an appropriate title.

An example output might be:

CustomerName	NotCollected
Jack Wright	2
Lin Cho	1
Santaya Yui	1

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[4]

(d) The shop uses a Database Management System (DBMS).

(i) One feature provided by a DBMS is a data dictionary.

The data dictionary stores the attribute names, table names, foreign keys and primary keys in the database.

Identify **three** other items stored in a data dictionary.

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(ii) The DBMS provides a developer interface.

Explain how a database designer can make use of the developer interface.

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2 A computer game developer stores data about a computer game in a database. The database has three entities:

- the users, who are called players
- the characters that users can create
- the groups that users can create and allow characters to join.

(a) Describe the relationships between the **three** entities.

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(b) The database is created with **four** tables:

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PLAYER(PlayerID, PlayerName, DateJoined)

CHARACTER(CharacterID, CharacterName, DateCreated, Level,
          NumberOfPoints, PlayerID)

GROUP(GroupID, GroupName, Category)

CHARACTER_GROUP(CharacterGroupID, CharacterID, GroupID,
                DateJoinedGroup)
```

Sample data for the table `CHARACTER` is given:

CharacterID	CharacterName	DateCreated	Level	NumberOfPoints	PlayerID
00123	Crystal	05/05/2025	5	35687	85AD
00124	Diamond	06/06/2025	4	21055	69CD
00125	Ruby	07/10/2025	1	1002	85AD

(i) The given Structured Query Language (SQL) script is written to define the table `CHARACTER`.

```
01   CREATE TABLE CHARACTER(
02         CharacterID INT,
03         CharacterName VARCHAR,
04         DateCreated DATE,
05         TheLevel INT,
06         NumberOfPoints INT,
07         PlayerID VARCHAR,
08         PRIMARY KEY CharacterID,
09         FOREIGN KEY(PlayerID)
10   );
```

The SQL script contains **four** errors.

Give the line number of each error and the corrected line.

error line 1 .....

corrected line .....

error line 2 .....

corrected line .....

error line 3 .....

corrected line .....

error line 4 .....

corrected line .....

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(ii) Write an SQL script to return each player's name and the number of characters each player has created.

The result needs to include a suitable field name for the number of characters.

Order the results in descending order of player ID.

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(c) Complete the description of database key terms by writing the missing words.

The ..... stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples of data that will be stored are ..... and ..... .

When designing a database, the ..... schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS).

A DBMS provides security features to protect the data. ..... are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data.

When creating a database, a ..... provides tools to create visual components for example forms and reports.

[5]

3 **(a)** State what is meant by the following terms in a relational database model.

Entity .....

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Primary key .....

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Referential integrity .....

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(b) Authentication is one method a Database Management System (DBMS) can use to improve the security of a database.

Describe **other** methods that a DBMS can use to improve the security of a database.

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(c) The following database table is not normalised.

StudentName	DateOfBirth	TutorGroup	Subject	SubjectCode
Yuwei Chen	01/09/2004	SMH	English, Maths, Computer Science	EN, MA, CS
Claudia Raj	23/02/2005	JMB	Maths, Physics, Art	MA, PY, AR
Aamil Akram	24/01/2005	KMB	Art, Design, English language	AR, DE, EN
Areeba Faraz	21/12/2004	SMH	English language, Chemistry, Design	EN, CH, DE

Explain how to modify the table to put it into First Normal Form (1NF).

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4 A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(a) Explain the benefits of using a relational database instead of a file-based approach.

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(b) The company decides which employees will work on each repair job. An employee can log into the database to access information about their repair jobs.

The database is normalised and includes these tables:

- `CUSTOMER` stores personal data about each customer
- `EMPLOYEE` stores personal data about each employee
- `LOGIN_DATA` stores the username and password for each employee
- `JOB` stores the data about each repair job
- `JOB_EMPLOYEE` stores the employees that are working on each repair job.

(i) Identify each relationship between the database tables **and** explain how each relationship can be implemented in the normalised database.

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(ii) The database also has the table `INVOICE` that stores data about each invoice that is sent to a customer.

Example data from the table `INVOICE` is given.

InvoiceID	DateSent	Amount	Paid	JobID
29262	12/12/2023	105.20	Y	221
26765	11/11/2023	200.00	Y	315
13290	02/01/2024	50.00	Y	315
34090	05/02/2024	25.95	N	569

Write a Structured Query Language (SQL) script to return the total amount of all the invoices sent in the year 2023 that have been paid.

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5 A company creates a relational database to store data about its customers.

The database, REVIEWS, stores data about the customers, products, complaints and staff.

The database has four tables:

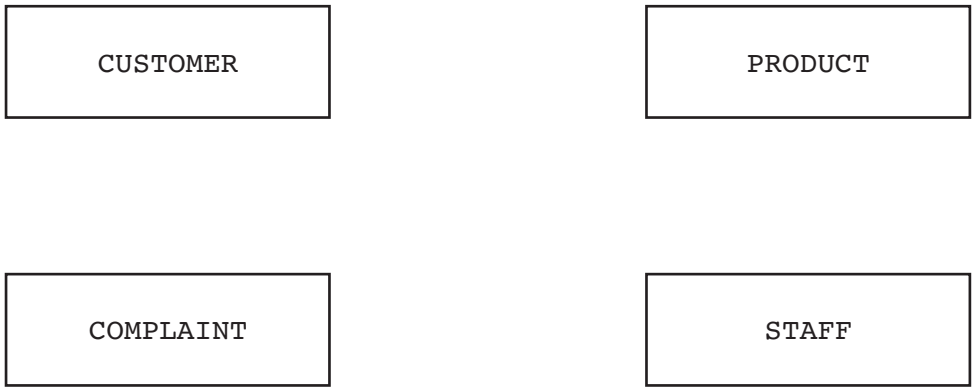
CUSTOMER(CustomerID, CustomerFirstName, CustomerLastName, CustomerEmail)

PRODUCT(ProductID, ProductName, ProductDetail, Price, Rating)

COMPLAINT(ComplaintID, ProductID, CustomerID, ComplaintDetails, StaffID)

STAFF(StaffID, StaffFirstName, StaffLastName, Department, RemoteWorker)

(a) Complete the entity-relationship (E-R) diagram for the database REVIEWS.



[3]

(b) Some example data from the `STAFF` table is shown.

StaffID	StaffFirstName	StaffLastName	Department	RemoteWorker
1	Ralph	Jura	E	Yes
2	Luca	Emcee	A	Yes
3	Darwin	Acula	F	No

Write a Structured Query Language (SQL) script to define the table `STAFF`.

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(c) Products are given a rating between 1 and 10 inclusive.

Write an SQL script to return only the `ProductID`, `ProductName` and `ComplaintDetails` for all products with a rating of 5 or less. The results need to be displayed in descending order of rating.

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(d) A Database Management System (DBMS) is used to maintain and manage the database.

Describe **two** ways in which the DBMS can be used to ensure the security of the customer data.

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Question 6 starts on the next page.

END OF QUESTION PAPER